

1. * A reference cathode is necessary because the absolute potential of a single electrode cannot be measured directly. Voltage can only be measured between two electrodes

* The Standard Hydrogen Electrode is used as a reference electrode because its potential is fixed at 0V

* Connecting the unknown half cell ~~to~~ to the standard Hydrogen electrode forms a complete circuit and produces a measurable voltage which is used to find the unknown electrode potential.

2. * Ionic movement in the electrolyte maintains electrical neutrality by balancing the ~~the~~ charge produced during oxidation and reduction reactions.

* This prevents charge buildup and allows continuous electron flow through the circuit.

3. * Transport numbers arise because cations and anions move at different speeds in a electrolyte solution. The faster moving ions carry a large share of the electric current

* The transport number represents the fraction of total current carried by a particular ion.

* It helps determine how the electric ~~field~~ current is distributed between cations and anions in the solution

4. * The Nernst-Planck Equation explains ion movement by combining both the ~~electric~~ effect of electric field and the concentration gradient.

* It shows that ions move due to electrical attraction and diffusion together. determining the total flux in the electrolyte

5. * The interfacial potential develops due to the transfer of metal ions between the electrode and the solution.
* Some metal atoms lose electrons and enter the solution as ions, creating charge separation at the interface.

* Equilibrium occurs when ionization and ion deposition happen at equal rates, maintaining a constant electrode potential without external current flow.

6. * An electrolyte is essential because it allows ion movement between the electrodes, completing the circuit.
* Without electrolyte, charge balance cannot be maintained, so no continuous current flows.

7. * Internal resistance increases because the electrolyte becomes weaker and unwanted materials build up inside the battery over time. This makes the movement of ions difficult.
* As a result, the battery gives less current, works less efficiently, and its performance decreases.

8. * Convection helps in transport by physically moving the electrolyte through stirring. This increases the movement of ions and reduces ion accumulation near the electrodes.

* As a result, mass transfer becomes faster, improving overall ion transport in the system.

9. Sol

The EMF of an electrochemical cell depends on concentration, temperature and pressure.

- * Change in reactant concentration alters electrode potential and usually decreases EMF.

- * Increase in temperature affects ion movement and reaction rate, causing EMF variation.

- * In gas electrode, higher pressure generally increases EMF.

Thus, variations in these factors cause fluctuations in the overall cell potential.

10. * Metal ions are reduced at the cathode and deposited as a metal coating.

- * The inert Platinum anode only supports oxidation reactions.

- * Uneven deposition occurs due to uneven current distribution and poor ion movement.

- * Ion depletion and increased resistance raise energy consumption during prolonged operation.

- * Proper electrode spacing, ~~stir~~ stirring and controlled current improve uniform coating and overall efficiency.

11. * A reversible cell operates slowly under equilibrium conditions with minimum energy loss and maximum efficiency.

- * An irreversible cell operates rapidly, causing heat loss, polarization and lower efficiency.

* In a reversible cell, the EMF is maximum and directly related to Gibbs free energy change.

* The relationship is given by:

$$\Delta G = -nFE_{\text{cell}}$$

* In an irreversible cell, energy losses reduce the observed EMF, so the electrical output is lower than the theoretical Gibbs free energy value.

12. The electrochemical cell works through redox reactions between zinc and copper electrodes.

* Zinc act as the anode and undergoes oxidation, releasing electrons.

* Copper act as the cathode where copper ions gain electrons and get deposited.

* Electrons flows through the external circuit from zinc to copper, producing electron

* The salt bridge containing potassium nitrate maintains electrical neutrality by allowing ion migration between two half cells.

* Over time, ion concentration changes near the junction create polarization and junction potential causing a gradual decrease in cell voltage

13. * The Helmholtz Double Layer Model assumes a rigid compact layer of oppositely charged ions near the electrode, similar to a parallel plate capacitor.

* The Gouy-Chapman model explains that ions are distributed diffusely due to thermal motion and electrostatic attraction

- * The Stern Model combines both theories by considering a compact layer near the electrode and diffuse layer beyond it.

Result: The Stern model best explains the electrical double layer because it accounts for both compact and diffuse ion distribution under different conditions

14. * The relationship between EMF and equilibrium constant is

$$E^{\circ} = \frac{0.0591}{n} \log K \quad \text{--- (1)}$$

Given: $E^{\circ} = 0.59V$, $n = 2$

Substitute above values in eqn (1)

$$0.59 = \frac{0.0591}{2} \log K$$

$$\log K = \frac{0.59 \times 2}{0.0591} \Rightarrow \log K = 20$$

$$\boxed{K = 10^{20}} \text{ equilibrium constant}$$

* The large values shows that the reaction strongly favours product formation at equilibrium.

15. * The potential gradient develops due to charge separation at the electrode-electrolyte interface, forming an electrical double layer.

* Potential drop is maximum near the electrode and decreases into the solution.

* Ion distribution controls charge separation at the interface

* Greater charge separation increases interfacial capacitance

* Thus, the potential gradient, charge distribution and capacitance are closely related in determining double layer behaviour.

16. * In a concentration cell, EMF is produced due to the difference in ion concentration between the two half cells.

* Using Nernst Equation

$$E_n = \frac{0.0591}{n} \log \frac{C_2}{C_1}$$

For concentrations 1M and 0.01M:

$$E = \frac{0.0591}{n} \log \frac{1}{0.01} = \frac{0.0591}{n} \log(100)$$

$$E = \frac{0.1182}{n}$$

$$[\log(100) = 2]$$

∴ EMF becomes positive, showing that current flows from the lower concentration half-cell to the higher concentration half-cell.

* The reaction quotient represents the concentration ratio which determines the direction of cell operation.

* The cell works until both concentration becomes equal and EMF becomes zero.

17. * Different electrode materials have different standard electrode potentials. The EMF of a cell depends on the difference between the electrode potential of the two electrodes.

* A larger difference in standard electrode potential produces higher EMF and greater energy efficient.

* $\therefore$ The electrode pair with the maximum potential difference is the most efficient.

18. * EMF represents the voltage supplied by the electrochemical cell. Higher EMF means the cell can produce more electrical energy.

* Electrical energy depends directly on the EMF and the amount of charge flowing through the cell.

19. * Activation loss occurs due to slow electrode reactions.

* Concentration loss develops from ion concentration imbalance near electrodes

* Resistance loss causes heat generation and reduces cell efficiency, making the cell irreversible

20. * A reversible cell operates under ideal conditions with constant voltage and maximum energy efficiency

* An irreversible cell shows voltage fluctuations and energy losses due to resistance and side reaction resulting in lower efficiency

21. * EMF changes when electrolyte concentration changes because electrode potential depends on ion concentration.

* Greater concentration difference gives higher EMF, while equal concentrations lower the EMF.

* $\therefore$ changing concentrations cause variable energy output and unstable cell performance

22. * Increase in temperature changes Gibbs free energy (ΔG), reducing the EMF of the cell when the system moves away from equilibrium.
- * Lower EMF results in decreased electrical energy output from the cell.
-

23. * In a real cell, heat generation and internal resistance produces entropy during current flow.
- * This entropy production causes energy dissipation, so part of the chemical energy is lost as heat instead of electrical work.
- * $\therefore$ practical efficiency becomes lower than the theoretical efficiency of an ideal reversible cell.
-

24. * Heating and voltage drop show energy loss inside the battery, indicating irreversible behaviours.
- * These inefficiencies reduce battery performance by lowering output voltage and decreasing overall energy efficiency.
-

25. * A secondary cell is the best choice for healthcare unit.
- * It is ~~not~~ rechargeable, reducing transport and maintenance needs.
- * Reversible reactions allow repeated use for long durations.
- * It provides stable and continuous energy output for medical equipment.
- * It has good energy efficient, high energy density and durable electrodes.

26. * Avoid overcharging and deep discharge to protect the electrodes.

* Use moderate charge-discharge cycles to maintain reversibility and battery life.

* Control temperature since extreme temperatures reduces efficiency.

* Prefer slow and balanced charging for stable performance.

* Regular monitoring helps reduce energy loss and maintain better capacity

27. * Electrode degradation reduces redox reaction efficiency, causing lower voltage output.

* Electrolyte combination decreases ion movement and affects cell reactions.

* Separator malfunction may block ion flow or cause internal short circuits, reducing performance

* Environmental exposure and poor assembly can increase corrosion and reaction instability.

* The most critical fault is usually electrode degradation because it directly affects electron transfer and overall energy output.

28. * Full discharge gives higher immediate energy output and longer driving range.

* Deep discharge increases electrode stress, reduces reversibility and shortens battery cycle life

- * Restricting discharge to about 60% improves battery lifespan and maintains better efficiency.
 - * It reduces usable energy per cycle but decreases degradation and replacement cost.
 - * $\therefore$ a moderate discharge strategy is optimal because it balances performance, efficiency and long-term durability.
-

29. The electrochemical cell qualifies as a reversible cell because:

- * It operates very slowly, maintaining equilibrium at all stages.
 - * Temperature and pressure remain constant, satisfying thermodynamic condition.
 - * The salt bridge allows smooth ion exchange and maintains charge balance.
 - * No heat lost occurs, so energy is not ~~lost~~ wasted.
 - * Maximum electrical work is obtained, which is the characteristic of a reversible electrochemical cell.
-

30. * Renewable source like solar and wind produce intermittent power, so efficient energy storage is necessary.

- * Lithium-ion batteries have limitations such as high cost, limited energy density and capacity degradation over time.
- * Large-scale storage increases infrastructure and maintenance expenses.

- * Frequent charge-discharge cycles under varying loads reduce battery life and efficiency.

- * $\therefore$ These limitations affect grid reliability and force engineers to consider more scalable, durable and cost-efficient storage technologies for renewable integration.

31. In a Daniell cell:

- * Increasing Zn^{2+} ion concentration shifts the cell reaction toward equilibrium.

- * According to Nernst equation, higher Zn^{2+} concentration reduces the EMF of the cell.

- * This happens because the tendency of Zn to oxidize decreases

- * Cu^{2+} concentration remains constant, so the change in potential is mainly due to Zn^{2+} ions

- * $\therefore$ the cell potential gradually decreases as Zn^{2+} concentration increases

32. * Conventional lithium-ion batteries use liquid electrolytes, which may leak, overheat or catch fire under extreme conditions.

- * Solid-state batteries use solid electrolytes, improving safety and reducing leakage risks.

- * ~~Thin~~ Solid-state ~~cell~~ technology provides higher energy density, giving longer operation with less weight.

* They perform better under extreme temperatures and mechanical stress, making them more reliable for aerospace applications.

* $\therefore$ solid-state batteries overcome many safety and stability limitations of traditional lithium-ion systems.

33. * variations in soil moisture and composition create differences in electrode potential on the same metal surface.

* This forms corrosion cells where one region act as the anode and another as the cathode.

* The anodic region undergoes corrosion, while cathodic region remains protected.

34. * corrosion occurs due to the presence of metal, electrolyte, oxygen and moisture forming an electrochemical cell.

* The ~~immersed~~ immersed region acts as the anode where metal dissolves, while the exposed region act as cathode, completing the corrosion process.

35. * Moisture form a thin electrolyte layer that allows ion flow and completes the corrosion cell.

* This enable continuous electron transfer, so corrosion happens faster in humid air.

* In dry air, no electrolyte layer forms, so corrosion is very slow or negligible.

36. * Acidic solution provides more H^+ ions, increasing electron transfer and accelerating metal dissolution.
* Hence, corrosion is faster with gas (H_2) evolution in acidic medium compared to neutral medium

37. * The corrosion is uniform corrosion where the entire surface is attacked evenly

* Coastal air provides moisture, oxygen and salt, forming a continuous electrolyte layer.

* This uniform exposure allows equal anodic and cathodic reactions across the surface, leading to uniform rusting

38. * Oxidation: Zinc loses electrons and forms Zn^{2+} ions

* Reduction: Another species gain those electrons.

* In corrosion, metal oxidation causes metal loss while reduction completes the electrochemical reaction

39. * Microscopic defects expose small areas of metal to electrolyte, forming localized anodic and cathodic regions

* Electron flow becomes concentrated at these exposed points, increasing reaction rate there.

* This localized current density accelerates corrosion kinetics at defect sites

40. * The growing oxide layer blocks electron transfer between metal and oxygen

* This slows down the reaction, so oxidation rate decreases over time

41. * Uneven soil condition creates differential aeration cells on the steel pipeline.

* Area with low oxygen/moisture imbalance become ~~cathodic~~ anodic, where iron oxidizes: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2e^-$
metal dissolves (corrosion occurs)

* Area with high oxygen become cathodic, where oxygen is reduced and no metal is lost.

* Electrons flow from anode to cathode through the metal, while ions move through soil moisture

* This continuous electron flows drives localized corrosion at anodic region, causing uneven pipe damage

42. Governing factors:

* Galvanic corrosion between stainless steel and carbon steel where carbon steel act as anode and corrodes faster

* Saline water increases conductivity, speeding up corrosion.

* Coating damage exposes, creating active corrosion sites.

* Flow stagnation ~~causes~~ causes oxygen difference, leading to localized attack near joints

Mitigation:

* Use galvanic series to select compatible materials or electrical insulation between metals

* Improve flow to avoid ~~ca~~ stagnation.

* Maintain coating and use corrosion protection methods (like cathodic protection)

43. Role of standard electrode potentials:

* Metal with more negative electrode potential are more reactive and act as anodes undergoing oxidation

* Metal with more positive electrode potential act as cathodes and protected:

Observation explanation:

* Zinc has a more negative electrode potential than copper, so it act as the anode and corrodes
($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$)

* copper has a more positive electrode ~~potential~~ potential, so it act as cathode and it is protected.

Conclusion:

* Metals with more negative standard electrode potential corrode first, so zinc corrodes while copper remains unaffected.

44. * From the galvanic series, steel is less noble than bronze. so, steel act as anode and corrodes while bronze act as cathode.

* Seawater act as electrolyte, enabling galvanic current ^{flow}.

* Large steel area + small bronze area increases current density on steel which leads to faster, localized corrosion at junctions.

* Oxygen variation further intensifies localized attack

* conclusion: steel preferentially corrodes near connections due to galvanic coupling and unfavourable area ratio.

45. * Under AC in Cu interconnects, electromigration, diffusion and electric field effects act together

* Electromigration (dominant): electrons flow ~~cause~~ cause Cu atom drift, leading to void/hillock

* Diffusion: atoms move from high to low concentration regions, especially at grain boundaries.

* AC effect: direction changes, but repeated cycle cause net atomic movement, so damage accumulates.

conclusion: Electromigration is the main cause, supported by diffusion and field effects, leading to gradual device degradation

46. * Zinc act as a sacrificial anode because it is more reactive than steel.

* Zinc undergoes oxidation: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$, protecting steel by supplying electrons.

* Steel becomes the cathode, so, it does not corrode.

* Continuous zinc loss shows it is actively protecting the hull.

* Effectiveness depend on reactive difference and good electrical contact between zinc and steel

Conclusion:

* Zinc corrodes preferentially, protecting steel through ~~sacr~~ sacrificial (cathodic) protection.

47. * stray currents create potential gradients along the pipeline, forming anode and cathode regions.

* ions in soil move under the electric field, supporting charge balance and maintaining the zones

* At anodic areas, metal dissolves ($\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$) which is high corrosion.

* Cathodic areas are protected due to reduction reactions.

* conclusion: Ionic transport driven by potential gradient concentrates corrosion at anode current-exit points, causing uneven damage.

48. * In saline soil, steel pipeline corrodes easily, so cathode protection is required.

* ~~sacrificial~~ sacrificial anode method (Zn/Mg):

i) simple, low cost but limited current output.

ii) It is suitable for short pipelines, ~~needs~~

iii) It needs frequent replacement.

* Impressed current method (ICCP):

i) uses external power, gives strong and adjustable protection.

ii) ~~eff~~ It is effective for long-distance pipelines but higher installation and monitoring cost

* For a long coastal crude-oil pipeline, current demand is high and uniform protection is needed.

conclusion: ICCP is more suitable due to better scalability, long-range protection and consistent control efficiency despite higher initial cost